

Project 13: Text to Speech

1. Background and Motivation

Text-to-Speech (TTS) aims to synthesize natural-sounding speech from text. Modern neural TTS systems typically decompose the problem into two stages: (i) an *acoustic model* that converts text into an intermediate representation such as mel-spectrograms, and (ii) a *vocoder* that converts these features into a time-domain waveform. Recent architectures (e.g., Tacotron2, FastSpeech2) and neural vocoders (HiFi-GAN) can produce high-quality speech, but training and evaluation can be challenging due to data quality, and compute constraints.

2. Problem Description

The primary goal is to build a TTS system that maps text input to audio output, and to evaluate it.

a) **Dataset:**

- Select a paired text–audio dataset (e.g., LJSpeech for single-speaker, LibriTTS for multi-speaker).

b) **Model Design (choose one main pipeline):**

- **Option A:** FastSpeech2 (text $\rightarrow$ mel) + HiFi-GAN (mel $\rightarrow$ waveform).
- **Option B:** Tacotron2 (autoregressive text $\rightarrow$ mel) + a neural vocoder (HiFi-GAN or WaveGlow).
- **Option C:** VITS (end-to-end text $\rightarrow$ waveform, with latent-variable modeling).

c) **Training and Inference:**

- Train the chosen model(s) and generate audio samples on a held-out test set.
- Report training budget (epochs/steps), batch size, and total GPU hours.
- Provide a small qualitative gallery (a few prompts) and discuss typical failure modes (mispronunciation, prosody, artifacts).

d) **Evaluation:**

- **Intelligibility proxy:** run an off-the-shelf ASR model on generated audio and report WER/CER (lower is better).
- **Spectral/prosody metrics:** report at least one objective metric such as MCD (mel cepstral distortion) or F_0 RMSE (or similarly motivated alternatives).
- **(Optional) Tokenizer:** character vs. phoneme-based tokens (or different normalization choices) and their impact on errors.

3. Expected Deliverables

Students working in groups (ideally 3 members) are expected to:

- produce a project report,
- deliver a group presentation,
- submit well-commented source code (training + inference + evaluation scripts).

4. Suggested Resources and References

- [1] Shen, J., Pang, R., Weiss, R. J., Schuster, M., Jaitly, N., Yang, Z., Chen, Z., Zhang, Y., Wang, Y., Skerry-Ryan, R., Saurous, R. A., Agiomyriannakis, Y., & Wu, Y. (2018). Natural TTS Synthesis by Conditioning WaveNet on Mel Spectrogram Predictions (Tacotron 2). In *ICASSP*. <https://arxiv.org/pdf/1712.05884>
- [2] Ren, Y., Hu, C., Tan, X., Qin, T., Zhao, S., Zhao, Z., & Liu, T.-Y. (2021). FastSpeech 2: Fast and High-Quality End-to-End Text to Speech. In *ICLR*. <https://arxiv.org/pdf/2006.04558>
- [3] Kim, J., Kong, J., & Son, J. (2021). Conditional Variational Autoencoder with Adversarial Learning for End-to-End Text-to-Speech (VITS). In *ICML*. <https://arxiv.org/pdf/2106.06103>
- [4] Ito, K. The LJ Speech Dataset. <https://keithito.com/LJ-Speech-Dataset/>